

1. Networking Hardware (Switches & APs)

- **Core Layer 3 Managed Switch:** Quantity **1**. Must support Gigabit throughput, VLAN tagging (802.1Q), and dynamic routing (RIP/OSPF) to act as the network gateway.
- **Access Layer 2 Managed Switches:** Quantity **3**. Each unit must be a 48-port Gigabit switch. These are required to handle VLAN 10 (Players), VLAN 20 (Admins), and VLAN 30 (Server Farm) traffic with QoS priority enabled.
- **Wireless Access Points:** Quantity **6**. Must be Wi-Fi 6 (802.11ax) compliant with PoE (Power over Ethernet) support to ensure sufficient coverage for 300+ spectator devices.

2. Cabling Infrastructure

- **Cat6 UTP Cable:** Replace all previous Cat5e specifications with **Cat6 UTP**. A sufficient quantity (calculated to cover approximately 100+ wired nodes and 6 AP locations) is required to maintain low-latency gaming traffic.
- **Patch Panels:** Quantity **5** (24-port, Cat6 compatible). Required to terminate the increased volume of physical cables from the gaming and admin zones.

3. Server & Management Hardware

- **Enterprise Server:** Quantity **1**. High-performance hardware required to host the Tournament Management System, Game Servers, VoIP, and Active Directory services defined in the Server Farm (VLAN 30).
- **Network Monitoring/Management Workstation:** Quantity **1**. Dedicated machine for running the Network Monitoring Dashboard (SNMP/real-time health monitoring).

Summary Procurement Table

Item	Specification	Quantity	Purpose
Core Switch	Layer 3 Managed	1	Inter-VLAN Routing & Traffic Filtering
Access Switches	48-Port L2 Managed	3	Wired device connectivity (VLAN 10/20)

Item	Specification	Quantity	Purpose
Wireless APs	Wi-Fi 6 (802.11ax)	6	Spectator Zone coverage (VLAN 40)
Cabling	Cat6 UTP	Bulk	High-speed, low-latency connectivity
Tournament Server	Rack/Tower Server	1	Hosting match instances & services